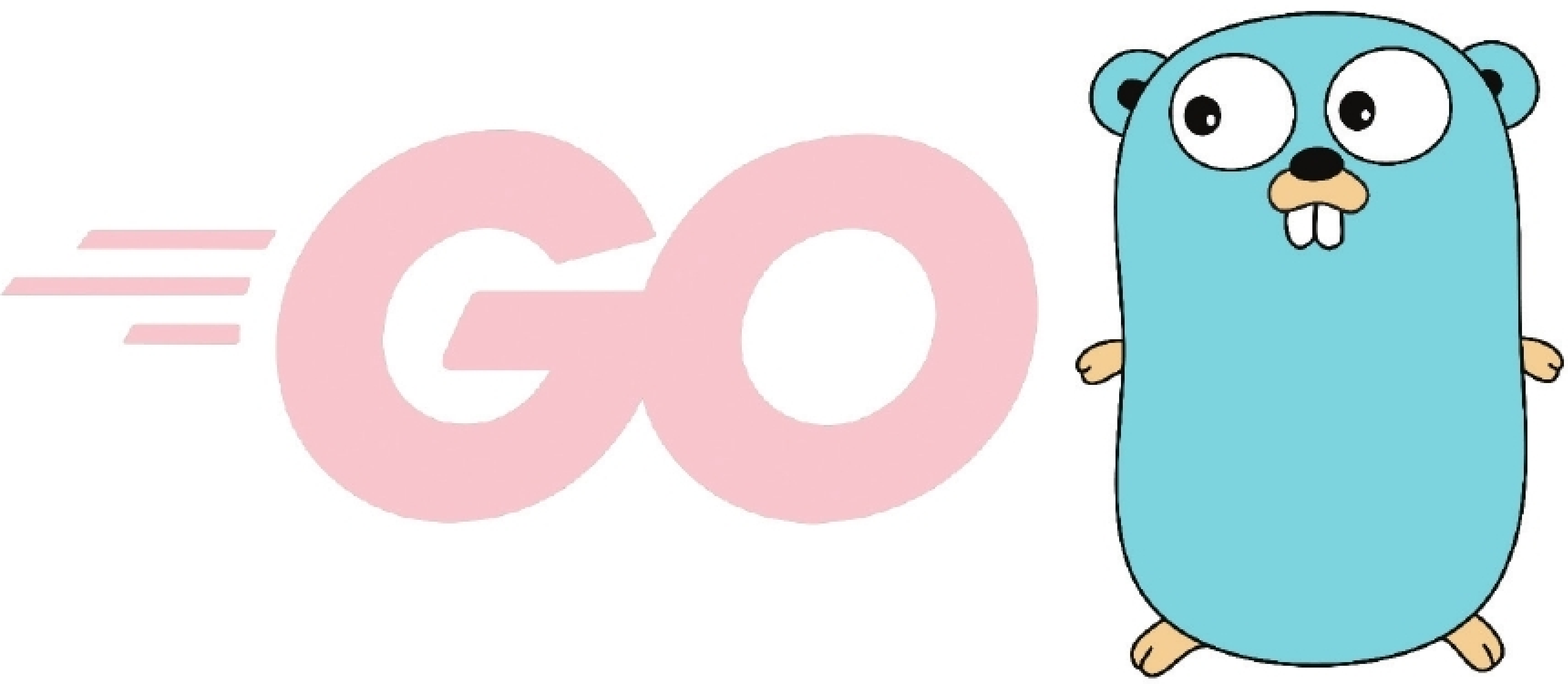
Go: Perhaps the Best Language for Building Scalable Code

This article on Go covers the reasons for learning it and some of its concepts. It also presents a sample program.



Go (also known as Golang) is a procedural programming language. It was developed in 2007 by Robert Griesemer, Rob Pike and Ken Thompson at Google, but it was only in 2009 that it was launched as an open source language. Since then it has became one of the most popular languages in the developer community. It provides a rich standard library, garbage collection and dynamic typing capability. Programs are assembled by using packages for the efficient management of dependencies, so that the code is easy to read, and the specifications are short.

Why learn Go?

So why exactly should someone learn Go? Golang was built by Google to help solve problems faced by developers at the company, where lots of big programs are built for server software that run on massive clusters. Before Go, Google was using languages like C++ and Java to solve problems, but these languages didn't really have the fluidity and ease of construction that was needed for problems at such scale. So Ken Thompson and Robert Greaser thought about building a new language that would be good for these kinds of problems and hence Golang was born. Basically, if you're trying to solve a problem of enormous scale or just need efficient and scalable code, go for Go.

To run the Go language in your own PC or laptop, you need a text editor and a compiler.

Text editor: This editor gives you a platform on which to write your source code. The following is a list of text editors:

- Windows Notepad
- OS Edit command
- Brief
- Epsilon
- vm or vi
- Emacs

Compiler: You can install the Go compiler from *https://golang.org/doc/install.* The extension of the source code file of Go language must be *.go*.

Writing your first program in Go

This must be typed in a text editor and saved as a file with a *.go* extension:

```
package main
import "fmt"
func main() {
```